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Social Separation in Monkeys as Viewed from Several Motivational Perspectives

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INTRODUCTION

Consider the following cases of human and nonhuman primate behavior. (a) When Billy was first admitted to the hospital 4 days ago, he was excitable, alternatively bubbling and crying but seldom quiet, perhaps not unusual for a 4-year-old in his first long stay away from his home and his parents. In the last 2 days, however, Billy's demeanor has changed considerably. He is no longer active; his behavior now ranges between silent pouts and quiet sobs. Once the darling of the nurses, he now refuses even ice cream. He spends most of his time lying on his back, staring at the ceiling. When his parents visit he becomes visibly sullen; tantrums often mar the visits. His parents are clearly worried as they contemplate the 8 weeks before Billy's treatment will be finished. They wonder if he will be his normal, outgoing self when he comes home again (Robertson and Bowlby, 1952). (b) The widow's children are worried. Since the death of her husband 7 months earlier, she has become more withdrawn, even though they now spend twice as much time with

The literature review in this paper is complete through 1977, reflecting the time when the chapter was written.

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her as they did before their father's passing. Yet her interest in the visits has been waning recently. She no longer cares if they bring their own children along—strange, since previously she had always taken a keen interest in her grandchildren. They fear that if her condition gets much worse she will have to be put in a nursing home. (c) A young chimpanzee, perhaps 2 or 3 years of age, remains at the periphery of a group of peers engaged in interactive play. The chimp is passive; he sits in a hunched-over posture. His eyes are red, betraying a lack of sleep. His few movements are slow and lethargic. Another chimp, slightly larger, approaches and grooms the youngster, then attempts to solicit reciprocal grooming. He does not respond. The older chimp is the infant's sister. Two weeks earlier, their mother was killed in an accident. Within a month, the young male will be dead as well (from Goodall, 1973). (d) Two separate groups of rhesus monkey adolescents have been reunited with each other after a 4-day period during which each group member had been housed alone. Yet the activities in the two group cages could scarcely be more different. In the first cage, the monkey adolescents alternate between bouts of intense reciprocal mounting and mock aggressive interactions. They are constantly in motion, yet somehow they manage to maintain some form of physical contact with each other most of the time. The monkeys in the second cage are of comparable age and sex distribution, but not nearly as active. Most clasp their own bodies instead of a partner. Two members of this second group stay in corners of the living unit; they have each rolled themselves up into a ball of fur. There is no play, no reciprocal mounting, no mock aggression in this group (Mineka *et al.*, 1981). (e) The once impressive-looking adult rhesus monkey male is now scarcely a shadow of his former self. Three months ago he had been unquestionably the dominant animal in the captive troop at the zoo. But then he was removed from the troop to be shipped to another primate facility, where he was destined to be the stud breeding male for a soon-to-be formed breeding colony. He has been living in quarantine for that 3-month period, housed individually in a small cage so that he can be rid of parasites and properly innoculated to permit safe entry into the new troop. He appears considerably the worse for wear. He has lost almost 40% of his previous body weight; skin once stretched over highly-developed muscles now hangs loosely from his sides. His fur is haggard. His new owners will be sadly surprised, for in spite of their fond plans, he will die before impregnating a single breeding colony female (Stephenson, 1977).

The above descriptions, objectively recorded examples of behavior exhibited by people and nonhuman primates in both laboratory and field environments, all share a common thread. Each subject's behaviors can be traced to the occurrence of a *social separation* from a conspecific individual with whom each subject had a prior, well-established social relationship. These descriptions document the potentially devastating effects that severing attachment bonds can have on complex social organisms. Yet the reports are hardly representative of all social separations that occur between individuals whose interactions with each other are mutually salient and significant. For each occurrence of a debilitating reaction to social separation in one individual, there exist numerous others that produce few, if any, problems for the separated partners.

For example, the young chimpanzee who reacted to the death of his mother with terminal depression had an older sister whose response to loss of mother indi-

cated little evidence of significant social debilitation (Goodall, 1973). The widow showing severe despondence in light of the loss of her husband differed considerably from the many middle-age individuals whose reactions to death of a spouse result in new, fruitful social activities. Many older adult rhesus monkey males have flourished as breeders after removal from competitive social groups. For every Billy there are dozens of Tommys and Marys who continue to be nurses' delights throughout their often painful hospital stays. These cases serve to remind us that there is enormous variability in primate reactions to social separations.

Such variability in reactions to social separation could result in large part from a variety of different factors. A considerably body of research has addressed some of these factors, often from differing theoretical viewpoints. In this chapter we will explore separation reactions in primates from some of these viewpoints. We will begin with a review of social separation research in macaque monkey subjects and will follow with a discussion of the different perspectives on separation phenomena that are provided by various levels of analysis. In addition, several theoretical interpretations of the monkey separation data will be presented, each derived from a different motivational viewpoint. Finally, we will suggest some research approaches that are directed toward resolution of some issues raised by the data and their various theoretical interpretations.

Study of Social Separation in Primates

Phenomena associated with the experimental interruption of attachment relationships have traditionally received widespread attention from those who study the social behavior of organisms. In research with mammalian subjects, separation of social partners from one another has been used to assess the strength of attachment bonds (e.g., Cairns, 1966), to discover mechanisms underlying social ontogeny (e.g., Hofer, 1975), to exaggerate the dynamics of existing social relationships (e.g., Hinde and Spencer-Booth, 1967), and to produce experimental psychopathology (e.g., Kaufman and Rosenblum, 1967). Some of the most extensive studies involving social separation have been carried out with nonhuman primate subjects, especially macaques such as rhesus monkeys.

Brief History of Macaque Mother–Infant Separation Studies

The rationale for the earliest systematic studies of separation in young monkeys can be traced to reports of clinical investigators examining young children who had been separated from their parents, usually because of hospitalization. These workers (e.g., Bowlby, 1960, 1973; Robertson and Bowlby, 1952; Spitz, 1946) found that for many children the loss of mother, even for relatively short periods of time, resulted in profound, long-lasting behavioral disorders. Not surprisingly, the first monkey separation studies, modeled after the human phenomena, were directed at disrupting the mother–infant bond.

The first published study of mother–infant separation in monkeys was performed at Wisconsin by Seay *et al.* (1962). These experimenters raised infant rhesus monkeys with their mothers for the first 6 months of life, separated the infants

from their mothers for a 3-week period during which the pairs could see and hear but not contact each other, and then reunited mothers with infants. Extensive observational data were collected during the 3-week period immediately preceding the separation, the 3-week period of separation, and the first 3 weeks of reunion. Seay *et al.* reported that the infants initially responded to separation with excessive vocalizations, disoriented locomotion, and repeated attempts to break through the barriers separating them from their mothers. Within a few days most infants changed considerably, becoming lethargic and socially withdrawn relative to their preseparation behaviors. When reunited, all mother–infant pairs showed enhanced attraction toward each other, but by 3 weeks their mutual interactions returned to preseparation base lines.

These findings bore striking similarities to reports in the human clinical literature. Bowlby (1960) had observed that many children separated from their mothers exhibited initial agitation, which he termed "protest," followed by weepiness and social withdrawal, which he termed "despair." Reunion, according to Bowlby, was often marked by "detachment," in which the infant expressed indifference and/or hostility toward its returned mother. Spitz (1946) had earlier reported somewhat similar reactions in younger children.

Subsequent monkey studies were more deliberately tailored for comparison with existing human data. For example, Hinde *et al.*, (1966) raised rhesus infants



Fig. 1. Typical "despair" behaviors of four rhesus monkey infants following separation from mother.

in pens containing mothers, peers, and an adult male for the first 6 months of life, removed the mothers from the pen for a 6-day period, and then returned the mothers to their infants. Hinde *et al.* reported that the infants showed clear-cut analogs to the "protest" and "despair" described by Bowlby (1960) for separated human children, although no evidence for "detachment" following reunion was found. Kaufman and Rosenblum (1967) raised pigtail macaque infants in similar social environments for the first 5–7 months of life, then removed each mother from the group pen for a 28-day period. A biphasic reaction ("agitation, depression") was exhibited by most infants, although some recovery was evident for all subjects by the third week of separation. Again, reunion produced little evidence of detachment.

In both studies, "protest" or "agitation" was characterized by hyperactivity and excessive vocalizations, while "despair" or "depression" was characterized by sharp decreases in play and other social activities and increases in passive, self-directed behavior. Figures 1 and 2 illustrate typical postures displayed by "depressed" monkeys. Reunion typically involved increases in mother-directed, contact-seeking and maintaining behaviors, hardly the sort of activity that might pass as "detachment." Typical reunion behavior is illustrated in Figure 3. Additional studies investigating mother–infant separation in other primate species have yielded similar findings; for example, Schlottmann and Seay (1972) in laboratory reared crab-eating macaques, and Goodall (1973) in ferally living chimpanzees.

However, some researchers studying other species of monkeys have not found such severe reactions to mother–infant separation. For example, Rosenblum and Kaufman (1968) separated bonnet macaque infants from their mothers. Although they employed the same paradigm as in their pigtail study mentioned above, they



Fig. 2. Close-up of separated infant.

discovered that the bonnet infants were scarcely affected by the separations. These infants showed no evidence of depression; rather, they redirected attachment behaviors to other adults remaining in the social group. Preston et al. (1970) separated patas monkey infants from mothers and found slight protest but no despair, with detachment-like behavior exhibited during reunion. Kaplan (1970) and Jones and Clark (1973) separated squirrel monkey infants from their mothers and reported well-defined agitation following separation, but again, no depression or despair.

Moreover, many investigators have found considerable within-species variability in the form, severity, and duration of reaction to separation. For example, Lewis *et al.* (1976) described data from five different studies of mother–infant separation of rhesus monkeys, all similar with respect to design, experimental procedures, and methods and forms of data collection. With respect to the results, however, the studies differed considerably. Variability in reaction to separation was enormous, and it could not readily be attributed to any procedural differences. At the very least, such results seriously question the existence of a "universal" response to maternal separation in this species. This is not entirely surprising, inasmuch as the human mother–infant literature reports variability in reaction to sep-

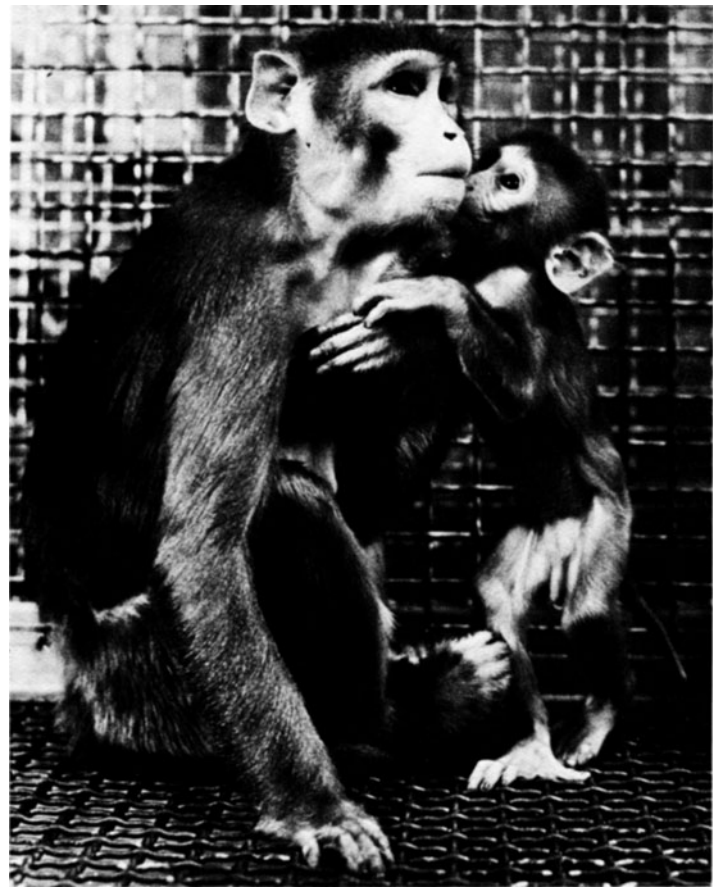


Fig. 3. Typical reunion behavior between monkey mother and infant.

aration of at least equivalent magnitude (e.g., Bowlby, 1973; Heinicke and Westheimer, 1966).

Variables Influencing Reaction to Maternal Separation

More recent studies of mother–infant separation in monkeys have examined a number of variables that might account for some of the differences in form of reaction seen across both subjects and situations. In addition to the species differences previously described, the possibility that there are age and gender differences within species has been investigated. Interestingly, these latter variables do not seem to be particularly important in determining an infant's response to separation from mother, at least within the first year of life (Hinde and Spencer-Booth, 1971; Rosevear, 1974; Suomi *et al.,* 1973). This is somewhat surprising in view of the consistently large age and sex differences in other aspects of monkey social development that clearly emerge within the first year (e.g., Hansen, 1966; Harlow and Lauersdorf, 1974; Harlow, Harlow, and Hansen, 1963; Hinde and White, 1974; Mitchell, 1968; Sackett and Ruppenthal, 1975).

Other studies have examined the duration and frequency of separation as potentially important variables in determining an individual's response to loss of mother. Most evidence suggests that the longer the period of separation, the more severe the short and long-term consequences for the infant will be, especially in its relations with its mother (Hinde and Spencer-Booth, 1971; Suomi, 1976). When separations have been maintained for more than 3 weeks, however, some "spontaneous recovery" has been reported for certain infants, particularly those that remain in the presence of socially responsive conspecifics (e.g., Rosenblum and Kaufman, 1968; Suomi *et al.,* 1973). Thus, severity of reaction to maternal separation is at best only partially a function of duration of separation. Few studies of mother–infant separation have examined the consequences of more than a single separation, but where this has been done the infant's reaction to subsequent separations from mother are at least as severe as its reaction to the first (Hinde and Spencer-Booth, 1971; Mitchell *et al.,* 1967; Suomi and Harlow, 1976). Apparently infants do not "adapt" to the separation procedure.

Thus, age or duration of separation does not account for most of the great variability seen in monkey infants' reactions to separation from mother. They may covary slightly with severity of reaction to separation across a large number of infants, but they do not explain the lion's share of individual differences often seen in subjects run in supposedly "identical" experimental protocols (e.g., Lewis *et al.,* 1976). A different experimental approach has focused on characteristics of an individual infant's social environment prior to, during, and following maternal separation and reunion. Here, investigators have tried to recreate social situations for infant monkeys thought by clinicians to be relevant for understanding human mother–infant separation reactions.

Preseparation Variables. Theorists of human attachment, from Freud to Bowlby, have emphasized that reactions to major social events such as separations from mother are in large part determined by one's previous history. In humans, differential separation reactions can often be traced to differential preseparation

histories (Bowlby, 1973; Heinicke and Westheimer, 1966). Specifically, the human data suggest that infants who are "insecurely attached" to their mothers (often due at least in part to previous separations) are far more likely to be adversely affected by separation from mother than are more "securely" attached infants. These data form the empirical foundation of many of Bowlby's theoretical ideas about separation.

The few studies with monkeys that have paid attention to an infant's social relationships with its mother and with other conspecifics prior to separation are largely consistent with each other, as well as with the human literature. For example, Kaufman and Rosenblum (1969) argued that the dramatic species differences in reaction to maternal separation shown by pigtail versus bonnet macaques could be traced in large part to species differences in how mothers rear their young. Pigtail mothers are possessive, seldom permitting their infants to interact with other conspecifics. In contrast, bonnet mothers are permissive, freely allowing their infants to form relationships with other adult females (such females are usually called "aunts" by primatologists). Thus, when pigtail infants lose their mothers, they have no one to turn to, whereas bonnet infants may lose a mother but as long as the "aunts" remain, they lose little mothering.

In an earlier study, Kaufman and Rosenblum (1967) noted that the only pigtail infant who was not depressed after maternal separation was the offspring of the dominant female in the infant's rearing group. This infant had spent more time interacting with other adult females prior to separation than had the other infants. In a more detailed study, Hinde and Spencer-Booth (1971) found that the more "dependent" an infant was toward its mother—that is, the more the infant strove to maintain contact with and proximity to its mother and the less often it left its mother and explored the environment—the more likely the infant was to exhibit severe reactions to maternal separation. These monkey data are thus consistent with the findings of several studies of human mother–infant separation regarding the importance of the preseparation relationship between mother and infant.

VARIATION IN SEPARATION ENVIRONMENT. The nature of the infant's social environment during the period of separation from mother also seems to influence the response to mother–infant separation. Hinde and his colleagues (Hinde and Davies, 1972) have demonstrated that an infant is more severely affected during separation when it is removed from the home environment while its mother stays than when the reverse occurs. However, following reunion the opposite is true—infants whose mothers are removed from the home environment take longer to return to preseparation patterns of behavior when they are reunited. Although these data are at variance with some human findings, Hinde (1976) has argued persuasively that the underlying mechanisms are probably similar for both groups of primates.

Another experimental approach has been to examine the effect of availability of substitute care for the infant monkey during separation from mother. As mentioned above, Kaufman and Rosenblum (1969) found that bonnet infants who were readily adopted by "aunts" after maternal separation showed no depressive reaction, whereas pigtail infants who were not adopted by "aunts" exhibited severe depression. Those few pigtails who did elicit maternal responses from other adult

females in their living groups did not become depressed. Similar findings have been reported by Schlottmann and Seay (1972) for crab-eating macaque infants and by Suomi *et al.*, (1976) for rhesus monkey infants. Again, there are parallel human data. Both Robertson and Robertson (1971) and Bowlby (1973) have reported that one of the best ways to minimize disruptive consequences of separation from mother is to provide the infant with a familiar maternal substitute.

Finally, some investigators have varied the access infants have to their mothers during separation. Suomi *et al.* (1976) found that when mothers and infants were separated by wide-gauge wire mesh partitions that permitted slight physical contact (e.g., reaching through the mesh), reactions to separation were negligible. However, when the same mothers and infants were separated by Plexiglas panels that permitted visual, auditory, and olfactory contact but prevented any tactile contact between mother and infant, protest–despair reactions followed. Previous research by Harlow and his colleagues (e.g., Seay and Harlow, 1965) had suggested that infants separated from their mothers by Plexiglas exhibited more severe reactions than those whose mothers were housed in a different room, out of sight and hearing of their infants. In addition, Kaufman (1973), reported data indicating that separated infants who have essentially "recovered" from loss of mother exhibit a dramatic reoccurrence of depressive symptoms when the mother is returned to their home environment but made inaccessible to physical contact. Thus, visual and auditory access to mother with physical contact denied is more stressful, in terms of severity of the infant's reactions, than is total sensory maternal deprivation. One might well conclude "out of sight, out of mind."

POSTSEPARATION VARIABLES. Most studies of mother–infant separation report little evidence of Bowlby's "detachment" following reunion. Instead, it has generally been found that immediately after reunion, mothers and infants interact with one another more intensely than they did immediately prior to separation. However, there is some variability with respect to how long this "enhanced" attraction is maintained. Some variability has been attributed to the nature of the reunion environment. Generally speaking, the more similar the reunion social environment is to the preseparation environment, the more rapidly mother and infant are likely to return to normal levels of behavior (Hinde, 1976; Suomi and Harlow, 1977).

Long-term effects of maternal separations have been reported by some investigators. For example, Hinde's group has found that following reunion, infants are more likely to try to maintain contact with their mother and less likely to leave her and interact with other conspecifics than are infants that were never separated from their mothers (Hinde, 1976). Moreover, such differences can be clearly detected 6 months after reunion, and in some cases they are still apparent 2 years later (Hinde and Spencer-Booth, 1971).

There can even be long-term consequences of maternal separation for infants when the separations are of extremely short duration. Mitchell *et al.* (1967) examined the reactions of juvenile monkeys to novel social stimuli and found that subjects who had been separated from their mothers every 2 weeks for *2-hr* periods during the first year of life were much more withdrawn when exposed to such stimuli than were monkeys who had not experienced such separations. Such data suggest that maternal separations that appear to be trivial at the time may in fact

have measurable effects that are not readily apparent until much later in a monkey's lifetime. However, maternal separation is not the only variable that produces behavioral disturbances later in life. Other studies have shown that nonseparation manipulations that elicit strong fear reactions in monkeys during the first year of life produce similar (although considerably weaker) long-term effects (e.g., Novak, 1973). Thus, maternal separation is only one of a family of "trauma-producing" events that may be associated with problems in coping when subjects are subsequently exposed to novel social environments.

SUMMARY OF MOTHER–INFANT SEPARATION DATA. Many studies have found that separation of infant monkeys from their mother can produce severe and often long-term disruption of social behavior; yet the degree of disruption varies considerably among individuals, both between and within species. Relatively little of the variance can be attributed to such global factors as age and gender of the subjects or duration and frequency of separation. Rather, factors prior to, during, and following separation all can contribute to the form of the overall reaction an individual monkey exhibits to separation from its mother. To date, there has not been much work directly investigating the relative contribution of these different sources of variance, nor has there been much effort designed to see how such factors might interact with each other in various mother–infant separations. We suspect that the interactions are substantial. At present it is exceedingly difficult to predict accurately the precise response of any given monkey infant to separation from its mother, even if one possesses detailed preseparation data and knowledge of parameters associated with the separation procedure to be performed.

PEER SEPARATION STUDIES IN RHESUS MONKEYS

During the past 5 years a growing body of research has been directed toward the study of social separation from nonmaternal objects of attachment. Monkeys and other mammals can and do form strong attachments to social objects other than their mothers, for example, surrogates (Harlow, 1958), peers (Harlow, 1969), and even television sets (Cairns, 1966). Researchers at Wisconsin have demonstrated that under certain circumstances separation from such attachment objects produces depressive symptoms in monkey subjects. For example, infants reared from birth without mothers but with peers exhibit protest–despair reactions when removed from their rearing group at any point during the first year of life (Suomi, 1971; Suomi and Harlow, 1975; Suomi *et al.*, 1970). Novak (1973) reported similar reactions to surrogate separation for monkey infants reared from birth on surrogates.

Furthermore, those variables that help determine the nature and course of mother–infant separations may have similar or identical effects for peer separations. For example, Suomi (1971) and Erwin *et al.* (1973) found that the quality of the attachment relationship and the overall behavioral repertoires of the subjects prior to peer separation were related to the severity of the subsequent separation, just as for maternal separations. Characteristics of the separation environment are probably as important in determining the course of peer separation as they are for maternal separation. Specifically, the more prolonged the separation and more socially restricted the environment when peer bonds are broken, the more likely will depression be the result (Suomi, 1973). Finally, it appears that the more

socially inadequate the reunion environment, the greater the probability of long-term depressive symptoms (Suomi and Harlow, 1975).

In sum, most variables found to influence reaction to separation from mother have similar effects in peer separations, at least among monkeys under 1 year of age. Indeed, qualitatively similar reactions to separation from mother and subsequent separation from peers can be readily elicited from the same subjects over a relatively short span of time (Suomi *et al.,* 1976). Not surprisingly the latter peer separation reactions are seldom as severe as those among monkey infants who have not been reared with mothers where the peer attachment bond is stronger. Clearly, severe separation reactions can be produced in young rhesus monkeys outside the confines of mother–infant relationships.

In several studies of peer separation, the monkeys have been separated from each other many times. In the first such study (Suomi *et al.,* 1970), monkeys were reared in groups of four without mothers from birth until 3 months of age. Between 3 and 9 months of age they were subjected to a total of 20 separations from each other, each 4 days long.

During every separation, group members were housed individually but could still see and hear each other. The monkeys reacted to each separation with initial agitation, followed by despairlike behavior. During each reunion the subjects spent most of their time clinging to one another. They showed little evidence of adaptation to the separation procedures, exhibiting about as severe reactions to the twentieth separation as they did to the first. Further, a cumulative effect of the 20 separations was almost total maturational arrest. After the last separation, when 9 months of age, the subjects were exhibiting essentially the same levels of all behaviors that they displayed at 3 months of age, prior to the first separation. This finding is noteworthy in view of the fact that under normal circumstances, the period from 3 to 9 months is characterized by more qualitative and quantitative behavioral change than during any other 6-month period in a rhesus monkey's lifetime (Suomi and Harlow, 1975).

Since the first study a number of additional experiments involving repetitive peer separations have been performed in rhesus monkeys under a year of age, and the results have been largely consistent with the original findings, particularly with respect to maturational arrest. For example, Seaman (1975) reared monkeys in groups of four from birth, without mothers, for the first 120 days of life, then subjected them to three 19-day separations, during which each monkey was individually housed. Subjects responded to each separation with protest–despair reactions, although by the third separation protest had diminished considerably. At the end of the third separation, when subjects were almost a year of age, they exhibited behavioral repertoires more similar to those of 120-day-old monkeys than those of nonseparated peer-reared control animals their own chronological age (Suomi *et al.,* 1978).

To summarize, repetitive peer separation in peer-reared rhesus monkeys under 1 year of age consistently produces depressivelike reactions to each separation, with a cumulative effect of maturational arrest. There appears to be less variability in reaction to repetitive peer separation procedures in young monkeys than in reaction to various forms of mother–infant separation. Possible implications of this finding will be discussed later in the chapter.

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Separation Studies in Older Monkeys

The separation data so far discussed have been limited to those involving monkeys under 1 year of age. Adolescent and adult monkeys also form social relationships with particular conspecifics, and these relationships can be experimentally interrupted. However, it is clear that for both humans and monkeys, separation may cause different reactions among infants, adolescents, and adults. Clinical reports of human cases of separation often describe quite different reactions to the same environmental events among individuals of different ages. Indeed, many of Spitz and Bowlby's pioneering studies prompted attention primarily because the reactions of infants to maternal separation seemed so striking at the time. For both humans and monkeys this is not surprising given that infants and adults have vastly different behavioral repertoires and often respond to a whole range of other events in their social environment in quite different fashions.

Relatively few studies have systematically examined the effects of separation from mother in adolescent and adult monkeys. This is understandable since ordinarily a young monkey will gradually separate itself from its mother (and vice versa), and by the time it has grown to adolescence the strength of the relationship has weakened considerably (Harlow and Harlow, 1965; Kaufman and Rosenblum, 1967). Also, for practical reasons, most laboratories performing mother–infant separation studies almost always separate offspring, particularly sons, from mothers long before the offspring have attained adolescence.

Nevertheless, several studies investigating the consequences of peer separation among adolescent and adult monkeys have been performed. These studies fall into four categories, each of which is described below.

Short-Term Peer Separations in Adolescent Monkeys. Studies involving short-term separation in adolescents have generally reported far less severe reactions than those shown by infants in comparable paradigms. For example, Bowden and McKinney (1972) separated recently formed peer pairs of 3-year-old subjects for a 2-week period, then reunited them. Separation was associated with mild protest but no despair, and reunion brought reciprocal mounting and rough play (these were all-male pairs). Although infants and adolescents possess different behavioral repertoires and on that basis alone might be expected to differ in form of reaction to separation, the fact that Bowden and McKinney's adolescents returned to preseparation base lines very shortly after reunion argues that their reactions were at best quite mild and thus different from the usual case for infant rhesus monkeys. McKinney *et al.* (1972), and Erwin *et al.* (1973) also reported only minor reactions to separations in adolescent monkeys.

Separations among Nuclear-Family-Reared Adolescents. Another possible reason for the relatively mild reactions to peer separation displayed by the juvenile monkeys described above is that since they had not spent a great deal of time together prior to separation, they may have been only mildly attached to each other. According to this interpretation, depressive reaction to separation does not depend on age; instead, subjects of any age that are sufficiently attached to an individual may exhibit a depressive reaction to loss of that individual. A recent study of Suomi *et al.* (1975) examined separation reactions in adolescent and young adult

rhesus monkeys housed from birth exclusively in complex social groups termed *nuclear families* (Harlow, 1971). Here, monkeys were reared in units containing four mother–father pairs and their offspring, so that each subject had access to other adults and to peers. When they were five years old, these monkeys were removed from their family units and housed in one of three conditions for a period of 120 days. Some were placed with familiar peers, others in groups containing strangers, and the remainder were housed individually. All of the group-housed adolescents exhibited mild reactions to separation from their families, with some aggression occurring in the group containing strangers. In sharp contrast, the individually housed subjects each fell into a deep depression following an initial period of protest. The depressive reactions of these monkeys could not be attributed entirely to the individual housing *per se,* because individual housing of juvenile monkeys in the separation studies previously discussed did not result in particularly adverse reactions. The phenomenon exhibited by the nuclear-family-reared monkeys would be described accurately as anaclitic depression had the subjects been infants, but a 5-year-old rhesus monkey is physically almost a full-fledged adult.

The results of this study show that many of the reactions to separation from mothers and/or peers characteristic of young infants can be seen in some physically mature monkeys under appropriate circumstances. Further, they suggest that some of the variables found to influence reaction to separation in infants may have similar effects among older monkeys.

Early Trauma and Later Reaction to Separation. Another general approach to the study of social separation in older monkey subjects has stemmed from the hypothesis that humans who have encountered traumatic social loss as children may be predisposed to depressive reaction to social loss as adults. Relevant human clinical data may be accurately described as mixed (Akiskal and McKinney, 1975). Researchers at Wisconsin have investigated this proposition in rhesus monkeys in three studies all somewhat similar in design. Subjects were either exposed to separation manipulations during the first year of life or were reared in intact social groups. All monkeys were then housed in social units until adolescence, at which time they were subjected to social separation.

The first study (Young *et al.,* 1973) employed a single 30-day peer separation as the infant manipulation and repetitive daily (23 hr/day) separations as the adolescent manipulation. Subjects reared in intact groups showed little adverse reactions to the adolescent separations, whereas, subjects who had experienced separation early in life exhibited depressivelike behaviors in reaction to the adolescent separations. A second study (McKinney *et al.,* 1973) compared the consequences of separation on 3-year-old monkeys that had been permanently separated from their social group early in life with those for previously nonseparated controls; it failed to detect any significant effects attributable to differential early rearing histories. A third study involved sequential separation of two groups of preadolescent monkeys, one of whose members had been subjected to sequential separation during the first year of life, the other remaining intact during a comparable period. The results indicated generally greater disruption by the separation in the former group of monkeys, but there were enormous individual differences within each group.

These findings suggest that although early traumatic social separation may in some ways predispose some monkeys' reactions to subsequent separations, early experience alone is a far from perfect predictor of reaction to later separation. Variables such as social experience since infancy, age at time of later separation, and form and severity of later trauma may also be important determinants.

LONG-TERM REPETITIVE PEER SEPARATIONS IN ADOLESCENTS. Results of two recently completed studies from Wisconsin indicate that separations repeated many times can profoundly change the behavioral repertoires of adolescent monkeys reared in peer groups since infancy. Chronic depressive behavior is one consequence of such manipulations (Mineka *et al.*, 1981). In both studies, subjects were raised from birth with peers but without mothers. During the first six months of life, monkeys in all groups but one were subjected to a limited number of repetitive peer separations. Then, after a variety of housing and testing experiences which largely kept the original rearing groups intact, all monkeys were later subjected to 16–24 weekly peer separations, each of 4 days duration. In one study, the separations began when the subjects were 16 months old; in the second study, separations began at 3½ years.

The results, similar in each study, were striking in three respects. First, most subjects exhibited severe protest–despair reactions to the first few separations in the series. Second, with subsequent separations all evidence of protest disappeared; subjects were showing "despair" reactions as soon as each separation began. Finally, after many separations, distinctions between separation behavior patterns and reunion behavior patterns became increasingly blurred. Reunion behaviors clearly changed more in the direction of "despair" behavior characteristic of separation than vice versa. Unlike younger monkeys, who usually respond to reunion with intense ventral clinging of the reunited mother or peer, and unlike the previously discussed adolescents, who responded to reunion with reciprocal mounting and mock aggressive activities, the present juveniles mostly sat next to each other passively during reunion periods, only occasionally clutching their own bodies or clinging to one another. They looked chronically depressed both during separation and reunion.

These new findings confirm that adolescent monkeys, like infants, can have their behavioral repertoires altered considerably by social separation manipulations. Further, it appears that the nature of reaction to social separation changes markedly as separations continue in these subjects. The theoretical import of these findings will be discussed at length later in the chapter.

SUMMARY OF SOCIAL SEPARATION FINDINGS IN MACAQUES

The results of the many studies of social separation that have been performed on macaque subjects can be summarized as follows:

1. Mother–infant separation usually produces intense reactions in infants, but there is enormous variability in the form of these reactions. Factors implicated by experimental studies are numerous; for example, species, age, gender, duration of separation, and nature of preseparation environment. However, interactions among these factors are presently not well understood, making actual prediction of any given individual's response to separation a hazardous venture. Long-term

effects of separation have been found in several studies; they seem to be more readily discernible when previously separated infants are observed in potentially stressful environments. There are almost no data regarding the consequences of repetitive maternal separation[1] or separation from mother in adolescents or young adults.

2. Peer separations in monkeys less than a year of age and reared exclusively in peer groups almost always yield intense separation reactions; when repeated they often yield maturational arrest. However, single or short-term repetitive peer separations in adolescent monkeys whose peer groups were first formed after 8 months of age, do not usually produce intense or adverse reactions.

3. Severe reactions to peer separation have been obtained in some adolescent and young adult monkeys, but other studies have reported at best mild separation reactions among adolescent subjects. Early experiences appear to be important; however, specifics regarding "sufficient" and "necessary" experiences in infancy required to establish such adolescent predispositions presently are not known. Current research suggests that multiple peer separations repeated many times during adolescence can produce marked behavioral changes in rhesus monkeys if the attachment bond is sufficiently strong.

Levels of Analysis

The preceding review indicates that social separation in monkeys often creates an intense affective/motivational state: the bonds broken by separation are affective, and individual differences in response to separation may well reflect motivational differences among individual subjects. As with any affective/motivational state, one can examine the phenomena associated with separation from several levels of analysis. Our analysis so far has been at a purely descriptive level, that is, correlating various behavior changes with certain social and environmental factors in the separation paradigm. Given that these behavioral data are not always very systematic across subjects and/or situations, it seems appropriate to examine the phenomena in question from other levels of analysis to help formulate explanatory principles as to why separation reactions take the form that they do.

For example, one might choose to examine response to separation at a *physiological* level of analysis. In such an approach, one might employ as dependent measures changes from preseparation base lines in sleep–wakefulness cycles, autonomic activity, patterns of eating, and/or neurotransmitter metabolism, either centrally or peripherally determined. Such data could then be used to assess individual differences in physiological reaction to separation among populations of subjects. They also could be examined for correlations with sociobehavioral changes associated with the separation manipulations. Alternatively, an investigator might compare reactions arising from separation manipulations with those resulting from other experimental manipulations known to have similar affective-motivational consequences. Similar physiological changes might be found for a variety of manip-

[1]We are currently running a study of repetitive mother–infant separations in which infants, beginning at 90 days of age, were subjected to a total of 16 4-day separations interspersed between 3-day reunions. Results to date suggest similar reactions as in parallel repetitive peer separation studies, but there are also intriguing influences related to the mothers' previous experiences.

ulations in subjects differing in age and/or previous history. On the other hand, the physiological substrate of affective/motivational states associated with social separation might be found to be unique.

More global levels of analysis of separation-related phenomena are also possible. Recently, an increasing amount of attention has been devoted to the *cognitive* variables involved in mediating a variety of affective/motivational states (e.g., Schachter and Singer, 1962; Seligman, 1975; Weiner, 1972, Zimbardo, 1969). In brief, the theme of such research is that the manner in which an organism perceives its environment and its relationship to the environment is an important determinant of its affective/motivational state. For example, with respect to separation-induced distress, it has been argued that the organism perceives its responses to loss of its mother or peer to be ineffective in bringing gratification or relief, comes to believe that it is "helpless," and therefore becomes depressed (Seligman, 1975). Studying cognitive changes that may accompany separation might well result in a more complete understanding of separation-induced phenomena.

Finally, we can ask *evolutionary/functional* questions about the adaptive value of such striking responses to separation. Some theorists, for example, Kaufman (1973), have proposed that immediately following separation, infant protest behaviors serve to draw attention to the fact that separation has occurred, possibly increasing the likelihood of reunion with the lost attachment object or of receiving substitute care from another conspecific, such as a maternal "aunt." Despair behaviors, on the other hand, which are characterized by quietness and inactivity, may serve to make the isolated organism less vulnerable to attack at a time when the probability of reunion is low. Bowlby (1969, 1973) has argued that one of the main functions of attachment behavior is to keep the child close to its attachment object, which provides a base of security from which it can explore its environment. When separation occurs this base of security is gone and normally low fear-eliciting stimuli now become terrifying. Other theorists (e.g., Solomon and Corbit, 1974) have speculated that the separation-induced behavioral states are the by-product of a perfectly normal *opponent-process* system in the brain which serves to bring an organism back to hedonic neutrality after a primary hedonic/motivational state has been stimulated, such as that associated with strong social attachment. This opponent-process approach to separation stresses that the separation-induced affective/motivational state is the manifestation of just one of many opponent-process systems whose function is to maintain the organism in a state of hedonic equilibrium.

In the sections that follow, we will examine five different approaches to the study of phenomena associated with social separation in nonhuman primates. We will pay particular attention to the degree to which the level of analysis employed aids understanding of the basis for the wide variability in separation reaction reported in the literature.

Physiological and Psychophysiological Correlates of Separation Phenomena

Most of the relevant studies regarding physiological consequences of social separation fall into two categories—those focusing on reactions to short-term sep-

arations (to which a "protest" behavioral reaction is most common), and those that have tracked individuals throughout the duration of more long-term separations, that is, those long enough to permit a "despair" behavioral reaction to develop in full.

One of the first studies in the former category was performed by Breese and his colleagues (1973). They separated six-month-old rhesus monkeys from their mothers and housed each in an individual cage for 6 days. The subjects were sacrificed at this point, and brain and peripheral tissue were assayed for levels of several neurotransmitters and enzymes. Relative to control subjects sacrificed immediately following separation from mother, these monkeys had significantly elevated levels of tyrosine hydroxylase, dopamine-B-hydroxylase, phenylethanolamine-*n*-methyl transferase, and choline acetylase in their adrenal tissue, as well as elevated tyrosine hydroxylase and choline acetylase in the superior cervical ganglion of the sympathetic nervous system and serotonin in the hypothalamus. During separation, they had been exhibiting behavioral protest, evidenced by excessive locomotion and vocalization. Breese *et al.* interpreted these findings as indicative of severe stress in the separated monkeys, grossly similar in form to neurochemical responses to stress produced in other animals using different techniques.

More recently, Levine and co-workers at Stanford have measured adrenocortical response in infant squirrel monkeys separated from their mothers for brief periods of time. Sharp increases in cortisol levels characterize most of these separations. For example, Hunt *et al.* (1977) removed mother–infant pairs from their social group for a 3-hr period, during which mothers and infants were either separated and housed individually or housed together in the same cage. Plasma cortisol levels were much more elevated in single-cage-housed infants than in those infants remaining with their mothers when removed from the group. Interestingly, Hunt *et al.* reported that the cortisol levels were not significantly correlated with the behavioral reactions of these subjects.

In a similar study, Levine *et al.* (1977) separated infant squirrel monkeys from their mothers by removing either the mother or the infant from the social group. Separations were either very brief (approximately 30 sec) or for 30 min. Relative to the briefly separated infants, those kept from their mothers for 30 min displayed enormous increases in levels of plasma cortisol.

These and other studies (e.g., Sassenrath, 1977) have consistently demonstrated that separation from mother is physiologically stressful for the monkey infant, although some of the physiological measures do not correlate significantly with the severity of behavioral reaction shown by each subject. The question of whether such reactions are unique to maternal separations (or any social separation, for that matter), or whether they are merely representative of a more general reaction to nonspecific stress remains to be determined. Here, data directly comparing reaction to separation with reaction to other well-known physiological stresses in same-aged monkeys with similar rearing histories would be most useful.

To date, most of the studies tracking physiological reactions to more long-term social separation have come from the primate laboratory at the University of Colorado Medical Center in Denver. Reite and co-workers (Reite *et al.*, 1974; Short and Reite, 1977) implanted minature transmitter devices in infant pigtail

macaques to monitor EEG, ECG, EOG, body temperature, and muscle activity prior to, during, and following maternal separation and reunion. Prior to separation there was little variability between subjects in heart rate, body temperature, or EEG. Immediately after separation, when subjects were exhibiting protest behaviors, heart rate and body temperature were elevated. During the first night of separation most infants also displayed a sharp decrease from preseparation levels both in amount of REM sleep and in heart rate during Stage 3 and Stage 4 sleep. After the second day of separation, intersubject variability on most physiological measures increased considerably. Some of those infants that exhibited despairlike behaviors (e.g., postural collapse) also had decreased body temperatures and heart rates, whereas the heart rates and body temperatures of most infants who did not exhibit despair returned to preseparation levels.

These data suggest that physiological reactions to separations of more than a few days may be as variable as behavioral reactions have proven to be. To date, there are not enough data to determine whether individual differences in physiological measures parallel individual differences in behavioral measures. We also do not know which variables are crucial in determining specific patterns of physiological reaction to long-term separation; indeed, such factors may be the same as those that determine the form of behavioral reaction.

Cognitive Approaches to Separation Phenomena

As discussed briefly above, cognitive theories of affect and motivation have recently received an increasing amount of attention (e.g., Irwin, 1971; Schachter and Singer, 1962; Seligman, 1975; Weiner, 1972; Zimbardo, 1969). Although differing in many respects, they share an underlying assumption that the cognitions an organism has about its environment and its relationship to the environment determine to a large extent its affective and motivational state. In these theories, cognitions precede and determine affect and motivation, although feedback from perception of one's affective or motivational state can, or course, subsequently affect and alter one's cognitions.

Despite the recent flurry of cognitive approaches to affective/motivational states, there has been surprisingly scant attention devoted to the importance of cognitive factors in response to separation. Littenberg *et al.* (1971) have shown that when a mother leaves her infant in a new and/or unpredictable manner, the response to the resulting separation is intensified. But little is really known about either the role of altered cognitions in mediating protest and/or despair responses to separation or, even more simply, the extent to which separation itself produces altered cognitions and what the nature of such alterations might be.

One cognitive approach directed toward separation has been the hypothesis of Seligman and his colleagues (1974, 1975; Seligman *et al.*, 1976) that *learned helplessness* underlies the despair/depression response. This approach to separation has occurred in the broader context of Seligman's elaboration of a learned helplessness model of depression. In brief, Seligman has proposed that *despair or depression* occurs when an organism learns or perceives that reinforcement and

relief occur independently of its responding, that is, when it learns or perceives that it is helpless to control its environment. In such a situation the organism's motivation to repond is greatly reduced, its ability to learn or perceive that a reinforcement has occurred because of its response is greatly impaired, and its affective state becomes one of sadness and depression. Seligman (1975) has suggested that one primary method of inducing such a sense of helplessness, and hence despair or depression, is by separating an organism from an attachment object, thereby removing a major figure through whom it gains a sense of control over its environment.

Helplessness, Depression, and Separation

The learned helplessness model of depression is a direct outgrowth of many experiments over the last 10 years that have studied the disruptive effects of uncontrollable aversive and appetitive events on a wide range of cognitive, social, and stress variables. As reviewed by Maier and Seligman (1976), exposure to uncontrollable as opposed to controllable events can lead to a reduced ability to learn new response–outcome contingencies, to lowered agression and loss of place in a dominance hierarchy, and to increased ulceration and lowered levels of norepinephrine. Impressed by the similarity of the cognitive, motivational, and affective changes produced by uncontrollable aversive events and those seen in depression, Seligman (1974, 1975) has proposed learned helplessness as a model of human reactive depression. More specifically, his basic premise is that perceived or actual loss of control over important life events is the major cause of depression, as it is of helplessness. Whereas the precipitating events in a depression may be loss of a loved one, loss of a job, or physical disease, to name a few, the real cause of the depression, according to helplessness theory, is that each of these events produces a belief in the person that he/she cannot control those things in his/her life which bring gratification (reinforcement) or relieve his/her suffering. Thus, the essence of the feeling produced by each of these events is helplessness. It is not the loss of reinforcement or gratification *per se* that produces the depression, but rather the sense of loss of control over a major source of reinforcement. Thus, the theory differs from that of Ferster (1974), who believes that loss of reinforcement *per se,* occurring for whatever reasons, is the major cause of depression. Specifically, with regard to separation, the learned helplessness hypothesis asserts that it is not the loss of the attachment object or other reinforcers *per se* that results in depressive responses to separation. Rather it is the perception of loss of control over the attachment object and/or the general loss of control over many important life events or reinforcers caused by the separation that results in depression.[2] When an organism perceives that it has no control over important life events, that is, that it is helpless,

[2]A recent reformulation of the learned helplessness model of human depression (Abramson *et al.,* 1978) now maintains that the type and severity of the depression will be a function of the attributions that the person makes for his/her helplessness. Because the utility of this model for understanding human depression remains to be determined and because we are not yet sure what implications it has for the study of depression in animals, we will confine our discussion to the original model (e.g., Seligman, 1974, 1975.)

this produces profound cognitive, motivational, and affective deficits: (1) The organism is retarded at learning in future situations that its responses can be effective in controlling the environment; (2) its motivation to respond in future situations is decreased because the incentive for responding stems from a belief that one's responses can be effective in producing gratification or relief; and (3) because of its sense of helplessness, its affect becomes characterized by sadness and passivity, and if the situation is severe enough, by depression. Under many circumstances, social separations create situations of helplessness for monkey subjects as indicated in the review presented above. If an individual loses a salient portion of its social environment via separation from an object(s) with whom previous bonds have been formed and no functional substitutes become available, the individual will have lost considerable control over its social options and/or opportunities. Moreover, if the separation and/or reunion environment is markedly different from the individual's preseparation environment, behavioral repertoires developed in the latter may not be entirely appropriate in the former, thus reducing the individual's social effectiveness and generating the perception of loss of competence and control on the part of the separated individual. These are precisely the cases of separation in which depressive reactions occur with greatest regularity, although at this point there is no direct evidence whether loss of control is, in fact, the mediating factor.

If a sense of helplessness is important in mediating separation-induced depression, then cognitive and motivational changes stemming from this sense should occur as a consequence of separation. The human separation literature clearly documents that such changes do result from at least some separations (Bowlby, 1973), although it is not entirely clear whether these result from the separation *per se* or rather from the deprived environment into which the children are often placed (Rutter, 1972). As yet no direct tests of motivational variables have been made in separated monkeys, but many of the symptoms previously discussed suggest that there are a variety of motivational deficits in the despair phase, for example, generally low activity levels and greatly reduced play and aggressive behaviors. In addition, the helplessness approach to separation suggests a search for cognitive deficits. The separated organism should be retarded in operant learning situations, that is, several repeated uncontrollable separation experiences should make it retarded at learning that it could have control over access to its attachment object. (See below for a discussion of the generality of such cognitive deficits.) To date we know of no evidence in monkey subjects bearing on these predictions.

The helplessness hypothesis also suggests several variables that should attenuate depressive responses to separation. First, monkeys able to control by their own responses when reunion occurs after some enforced period of separation should show a markedly attenuated response to separation as compared to yoked monkeys who have experienced the same separation but have no control over reunion, that is, for whom reunion occurs at the same time but noncontingently. Second, having the separation occur in an environment where there are still multiple ways of exerting a sense of control should also markedly attenuate the depression response. For example, less depression should occur when an organism is separated into an environment where it has potential control over access to other attachment objects and/or various manipulanda for eating and playing than when separated into an envi-

ronment where the same sources of reinforcement are available noncontingently. Such a prediction is generally supported by a wealth of human and monkey separation data previously discussed, although none of those studies were designed with the helplessness hypothesis in mind, and so nothing definitive can presently be concluded about the importance of control *per se.*

The learned helplessness approach to separation also suggests a number of factors that may be involved in predisposing an organism to a particularly severe, or a particularly mild, response to separation. While not denying the importance of biological or genetic factors in predisposing people to depression, the model emphasizes the importance of early experience with control over important life events as the major factor in predisposing a person to depression. Very early in the series of learned helplessness experiments in animals, it was noted that the disruptive effects of inescapable shock could be prevented by prior exposure to escapable shock (Seligman and Maier, 1967). Thus, dogs given some escape/avoidance training before exposure to inescapable shocks are not subsequently helpless, in contrast to dogs without such previous experience. Seligman *et al.* (1976) further hypothesized that the immunization experience provided by the prior exposure to escapable shock need not be this specific; a prior history of control over any important life events may be sufficient for immunization against the disruptive effects of exposure to uncontrollable events. In support of this view isolation-reared or laboratory-raised dogs are more susceptible to the effects of inescapable shock than are dogs of unknown history (Lessac and Solomon, 1969; Seligman and Groves, 1970). The isolation-or laboratory-reared dogs have presumably had less of a history of control over their environment because of more restricted opportunities for such control. Recently, Williams and Maier (1977) more directly demonstrated transituational immunization in rats by showing that learning one escape response (e.g., wheel-turning) immunized the rats against the disruptive effects of inescapable shock when the rats were later tested on a topographically different response (e.g., shuttling). In addition, Joffe *et al.* (1973) found that rats raised in a "controllable" environment, where food, water, and light were only available by lever pressing, were less emotional in an open field test later in life than rats raised in a noncontingent environment. Thus, there seems to be some support for Seligman's notions on immunization.

With regard to depression, this theory predicts that individuals with a prior history of control over major life events will be less susceptible to depression following a given traumatic event than will other individuals who have a history of relative lack of control over their environment. If one assumes that separation from an attachment figure is an uncontrollable traumatic event, then a history of repeated separations involves a series of exposures to uncontrollable traumatic events. From the helplessness standpoint then, we would expect to find separation to be a depression-inducing manipulation, as it clearly is, and also to find that monkeys exposed to repeated separations would be particularly susceptible to depression later in life. This prediction has been confirmed by some studies but not by others (see p. 555). As yet, however, there are no data regarding the importance of a previous history of control *per se,* either over separation or over other important life events, in determining later response to separation.

At first glance the helplessness approach to separation does not appear to explain the highly agitated and disturbed behavior almost invariably seen following most separations, even many that do not subsequently result in depression. However, a recent integration of helplessness thoery and reactance theory proposed by Wortman and Brehm (1975) offers a possible resolution. Reactance theory asserts that when an organism loses control and perceives that its behavioral freedom is being threatened, it becomes motivationally aroused (reactance) and tries to restore its freedom. For example, during the initial exposure to inescapable shock, dogs commonly make vigorous attempts to escape. The amount of motivational arousal or reactance that the organism experiences is partly a function of "the expectation that he possesses freedom to begin with" and "the importance of the freedom threatened" (1975, p. 283). The protest behaviors seen as an initial response to many separations are consistent with Wortman and Brehm's description of the state of reactance. Furthermore, reactance theory predicts the data, discussed above, that indicate a dramatic decline in the protest phase with repeated separations. With repeated uncontrollable separations, the organism's expectation that it possesses freedom must begin to decline, and with the decline of that expectation its reactance (protest) should also decline.

Since the reactance phase as a response to loss of control is opposite to the helpless phase, Wortman and Brehm have concentrated on integrating the two theories. They contend that the state of reactance or motivational arousal lasts for a finite period of time. With extended exposure to uncontrollable outcomes, for example, repeated separations, the organism becomes convinced that it cannot control its outcomes and gives up trying, that is, the organism becomes helpless in Seligman's terms. Recently Roth and Kubal (1975) have provided support for this integrative model by demonstrating both reactance and helplessness in the same study. Brief treatment with uncontrollability produced reactance whereas prolonged treatment with the same uncontrollable outcomes produced helplessness. If separation from an attachment object is seen as a highly aversive uncontrollable event, then the initial response should be akin to reactance—protest—and the final response should be akin to helplessness—despair. With repeated separations, as we have seen earlier, the protest phase (reactance) should virtually disappear and the despair phase should become more prominent.

SUMMARY

Overall, the learned helplessness approach to separation appears to be useful for several reasons. First, it provides a new framework in which to study the importance of cognitive variables in the response to separation. This emphasis on the role that cognitions—especially those concerning response-outcome contingencies—can play in mediating affective/motivational states is part of a more general recent trend of bringing "the mind" back into experimental psychology. Second, the helplessness approach makes some specific predictions that differ from those of other theorists (e.g., Ferster) as to what factors may attenuate or ameliorate separation-

induced depression. Some of these predictions, for example, may be tested by comparing subjects who do and who do not have control over reunion with their attachment figure, or by comparing subjects who do and who do not have control over various sources of reinforcement in their pre- and/or postseparation environment. Third, this approach makes interesting and testable predictions about what kinds of early experiences predispose an organism to separation-induced depression later in life. For example, subjects with an extensive history of exposure to noncontingent positive or negative events should be more prone to such depression than subjects with a history of control over such events. Finally, it provides a plausible explanation for the common occurrence of protest immediately following social separation, if one allows for some integration of helplessness and reactance theories.

Functional Analyses of Separation Phenomena

As an alternative to behavioral, physiological, or cognitive analyses of phenomena produced by social separation, a number of theorists have instead focused on postulated *functions* of particular response patterns observed in reaction to separation. Such functions are generally couched in terms of larger theories dealing with a wide range of phenomena, not just those associated with social separation. In this section we will examine three such approaches: Bowlby's attachment–object-loss approach, Kaufman's conservation–withdrawal analysis, and Solomon's opponent-process theory of motivation.

Bowlby's Attachment–Object-Loss Approach to Separation

John Bowlby (1969, 1973) has developed a comprehensive theory of human and animal attachment behavior heavily based upon principles derived from the diverse fields of ethology, psychoanalysis, and social learning. In his approach, Bowlby devotes considerable attention to the function that attachment behavior serves but only indirectly addresses the issue of the function that particular reactions to separation from an attachment object might serve. In brief, he asserts that attachment behaviors keep the organism near a base of security that can reduce the impact of fear-eliciting stimuli. Separation produces a universal reaction of upset and distress because the secure base that usually alleviates fear and distress is removed. Repeated experience with separation can lead to the organism developing a fear of separation, with the resulting behavior characterized as "anxious attachment." Details of these points follow.

Function of Attachment Behavior. Bowlby argues that one of the most important functions a maternal attachment object serves for a developing infant is to provide a source of security—a base from which to explore the environment, and a safe refuge to return to in the presence of fear stimuli. This latter role is seen is especially important because the infant, even in "normal" environments, is often exposed to potentially frightening stimuli. Some stimulus patterns, for example, rapidly approaching objects, loud noises, and darkness, appear to be unlearned

elicitors of fear in young primate infants, human and nonhuman. Other fears develop as the infant matures, for example, fear of strangers, which appears in rhesus monkeys around 2–3 months of age and in humans during the last third of the first year of life. Still other fears are acquired through conditioning processes or by observational learning. Bowlby argues that it is highly adaptive for infants and young children to respond with fear to certain stimuli, and there is nothing intrinsically wrong or abnormal about such a response.

According to Bowlby,

> a cornerstone of the present argument is that . . . (in) behavior indicative of fear are to be found forms of behavior that have at least three different kinds of predictable outcome: (a) immobility, (b) increased distance from some type of object (the fear stimulus) and (c) increased proximity to another type of object (the attachment object) . . . Both [of the latter] behavior(s) do not always occur. Yet they occur together with sufficient frequency to enable us to take the combination for granted . . . When a child is afraid of a barking dog, we expect him not only to withdraw *from* the barking dog but to retreat *towards* a parent figure. (Bowlby, 1973, p. 89)

Thus, the presence of an attachment figure provides the infant with a means to escape from or reduce the frightfulness of fear stimuli.

It follows that the removal of an attachment object results in the infant losing its refuge from potential fear stimuli. Put another way, stimuli that are frightening to an infant in the presence of an attachment object become even more frightening when the infant is alone. Bowlby calls this a "compound situation." When one is alone, ordinary stimuli may become frightening, and normally fear-provoking stimuli may become terrifying. Under such circumstances, being alone can be very unplesant—in fact, Bowlby presents convincing data in both young humans and young macaques that being away from one's attachment object is in itself fear-producing.

In view of these data, Bowlby has argued that there is a "universal" reaction—distress—to separation from an attachment object (Bowlby, 1960, 1973). In extreme cases, that is, in circumstances when separations are prolonged, the distress takes the form of the protest, despair, and (with reunion) detachment sequence described earlier. Thus, proximity to an attachment object is necessary to feel secure and unafraid, and there are good adaptive functional reasons why that should be the case. It follows logically that separation from the attachment object should be sufficient to induce severe distress.

SEPARATION ANXIETY AND ANXIOUS ATTACHMENT. Because separation from an attachment object produces distress in and of itself and has the potential to make moderately fear-provoking stimuli considerably more terrifying, individuals who have undergone particularly unpleasant separation experiences are not likely to forget them. Rather, according to Bowlby, such individuals may develop a fear that they may, at some time in the future, be separated again. Bowlby calls this "fear of separation" (Bowlby, 1973, Chap. 12); it is closely related to what others such as Freud (1917/1953) have termed "separation anxiety." Fear of separation develops under conditions in which (a) the individual has had previous unpleasant separation experiences, and (b) the individual perceives, for real or "imaginary" reasons, that separation is likely to occur again in the near future.

One consequence of fear of separation in young humans, according to Bowlby (1973), is the "anxious attachment" exhibited by some infants and/or young children. Anxiously attached children are reluctant to leave the sight of their attachment object for even short periods of time, they tend to exhibit excessive proximity seeking and maintaining behavior, and they are more prone to react severely to subsequent separations than are "securely" attached children. However, fear of separation is not the only cause of anxious atachment—it is influenced by numerous factors, including constitutional factors of the infant, care-taking characteristics of the mother, and various aspects of the social environment (cf. Bowlby, 1973, pp. 212–236). Unfortunately, at present the relative contribution of each factor and possible interactions have not been specified.

APPLICATION OF BOWLBY'S APPROACH TO MACAQUE SEPARATION DATA. In general, Bowlby's theoretical considerations are strikingly consistent with much of the macaque separation data. This is not surprising, because Bowlby has based many of his formulations on the monkey data from the Harlow and Hinde primate laboratories (Bowlby, 1975). Some time ago, Bowlby (1960) proposed that severe reaction to separation was biphasic during separation, and certainly biphasic protest–despair reactions to maternal separation have been reported in a number of macaque species. However, little documentation of detachment in macaques following reunion exists. Thus, Bowlby's contention that detachment is a basic component of the separation reaction does not appear to generalize widely to nonhuman primates.

Bowlby's theory flatly states that insecurely attached infants, that is, those that exhibit "anxious attachment," should be the most adversely affected by long-term separations. In agreement with this, Hinde and Spencer-Booth's (1971) data indicate that rhesus infants who appeared to be most anxiously attached, that is, those who spent the most time trying to maintain contact with the mother prior to separation, were most likely to show tension in the relationship upon reunion. Other monkey data (e.g., Kaufman and Rosenblum, 1967) report essentially the same finding. However, the exact contribution of previous experience with separation in producing anxious attachment is not clear at this time because little work has been done with repetitive maternal separations. As mentioned previously (see Footnote 1), we are currently running a repetitive maternal separation study in which infants are separated for 4-day periods interspersed between 3-day reunions. Although the data analyses have not yet been completed, it is clear that relative to nonseparated controls, repetitively separated infants display unusually high levels of mother-directed behaviors and virtually no exploration of the environment. This "anxious attachment" is apparent from the first reunion periods, it becomes more pronounced with repeated separations, and it continues to be displayed by infants long after the separations have been terminated. Interestingly, individual differences in infant's reactions seem to be related to the *mother's* prior rearing and separation histories.

Another point in Bowlby's theory that is consistent with the primate literature is that a primary function of an attachment figure is to provide a security base. Harlow and Zimmermann (1959), for example, found that infant monkeys retreat to a cloth surrogate when faced with frightening stimuli. Contact with the surro-

gate appears to produce considerable relief and relaxation as compared to the fearful state that monkeys reared without a surrogate or with a wire surrogate remain in under comparable circumstances. Bowlby's theory also predicts that those infants who stay with maternal substitutes (who can provide an alternative base of security) during separation from mother will exhibit trivial reactions compared to those infants without maternal substitutes. Again, the monkey data are largely consistent with the prediction, as are data on human children (Robertson and Robertson, 1971).

Finally, the theory provides a reasonable explanation as to why apparently minor separations, for example, of less than 2-hr duration, are capable of producing long-term sensitization to novel stimuli (Mitchell *et al.*, 1967). Such short-term separations may be sufficient to establish a fear of separation.

All in all, most findings in the monkey maternal separation literature are consistent with Bowlby's attachment–object-loss approach to separation phenomena, if one ignores the basically negative findings regarding detachment in monkeys reunited with their separated attachment objects. The thoery also generates numerous interesting and testable predictions as to which factors will minimize distress upon separation and what the effects will be of different kinds of separation experiences on the dynamics of an attachment relationship.

Kaufman's Conservation–Withdrawal Approach to Separation

Kaufman (1973; Kaufman and Rosenblum, 1967) has a different approach to the possible functions of particular reactions to separation. He has suggested that a biphasic protest–despair reaction to maternal separation is just one example of many biphasic responses that organisms manifest to intensely stressful situations: first, a *flight–fight* reaction, followed by a phase of *conservation–withdrawal* when the first reaction threatens to exhaust the organism. This theory proposes that our nervous systems are organized so as to mediate this type of biphasic response to any "mounting need" (Kaufman and Rosenblum, 1967, p. 667). Specifically with regard to separation, protest is seen as an example of the flight–fight response and despair as an example of conservation–withdrawal.

Kaufman speculates extensively about the adaptive nature of this biphasic pattern of response to separation. The protest phase mobilizes the organism's energy and resources after a separation so as to increase the likelihood of regaining contact with the last attachment object. For example, it can be argued that the increased locomotion shown by infants during protest serves the function of increasing the likelihood of an infant finding its lost mother and the excessive vocalizations increase the likelihood that the mother will find her lost infant.

The protest phase, however, cannot persist indefinitely without causing complete exhaustion. Furthermore, these behaviors are eventually likely to attract attention and/or aggression from both conspecifics and a variety of predators. Hence, the second phase of despair or depression conserves the infant's energy at a time when the probability of reunion is likely to be low. Kaufman argues that because this phase is characterized by inactivity and huddling postures ("postural collapse") it serves to keep the separated organism from attracting the attention of

predators and other potentially dangerous agents. In addition, he speculates that the "dejected baby" posture may serve a communicative role in some species, increasing the likelihood that a conspecific may take over the protective maternal role.

Kaufman emphasizes that this second phase of despair is not a manifestation of massive fatigue resulting from the protest phase but rather occurs *to prevent* such fatigue from setting in. In the despair phase the organism can respond appropriately to situations calling for energy, for example, being the object of an aggressive attack. Furthermore, when reunion occurs, the infant typically rapidly recovers to baseline preseparation activity levels. This would not occur if the despair phase were truly a fatigue state.

A third phase—spontaneous recovery from depression—is sometimes seen in monkeys who have been separated for long periods. Kaufman suggests that this recovery can occur in infant monkeys because of their relatively high degree of locomotor ability after the first few weeks of life (as compared, say, to a human infant). Such recovery is obviously adaptive in allowing the infant to find new sources of attachment and begin to resume more normal courses of development. He argues that in the human case there may be no recovery if the separation is continued (e.g., Spitz, 1946) because of the human infant's lack of locomotor ability and its greater degree of dependence on its mother.

Kaufman's conservation-withdrawal approach to separation is consistent with a number of neurobiological theories that emphasize how the organization of the nervous system has evolved so as to mediate two opposing responses to stressful states. For example, in Hess's terms (1954) the sympathetic nervous system underlies and mediates the "ergotrophic" or mobilizing system—the flight-fight response or, in the case of separation, protest. The parasympathetic nervous system underlies and mediates the "trophotrophic" or energy-conserving system—the withdrawal response of despair or depression after separation. Kaufman also discusses other biphasic response theories relevant to his theory of protest-despair, for example, Riss and Scalia's (1967) comfort-distress dichotomy, Engel's (1962) facilitation-suppression of distress reactions, and Schneirla's (1959) approach-withdrawal responses. The physiological correlates of protest and despair discussed earlier (pp. 558–560) are generally consistent with these biphasic response theories.

Thus, Kaufman provides an analysis of separation phenomena that ascribes plausible functions to protest and despair patterns of behavior. These functions are consistent with both evolutionary theory and present knowledge of physiological variables, and they can be studied at both behavioral and physiological levels of analysis. Kaufman has also addressed the question of the nature of the cognitive and subjective concomitants of the protest-despair response. He maintains that the protest phase is akin to a state of anxiety and the despair phase to a state of helplessness and depression. However, as yet these cognitive/subjective states in monkeys are only indexed by the distinctive and characteristic behavior patterns that occur in each, that is, excessive agitation, vocalizations and anguished facial expressions during protest, and withdrawal, inactivity, collapsed posture, and grieflike facial expressions during despair. Until the cognitive correlates (helplessness) have

been more adequately operationalized, it is not clear how useful this aspect of Kaufman's theory will be for generating future research ideas.

Solomon and Corbit's Opponent-Process Approach to Separation

Solomon and Corbit (1973, 1974; Solomon, 1977) have recently proposed an opponent-process model of acquired motivation that provides another framework in which to study the response to separation from attachment objects. Like Kaufman (1973; Kaufman and Rosenblum, 1967), they stress the adaptive function that a profound response to separation serves, but there is a major difference in emphasis as to what this function is. Kaufman emphasizes that protest behaviors increase the probability of reunion whereas despair behaviors conserve energy and decrease the probability of attack when the probability of reunion is low. Solomon and Corbit, in contrast, emphasize that the response to separation is a typical by-product of the termination of *any* intense affective/motivational state such as the one created by a strong attachment relationship. The function that all such "opponent-processes" serve is to bring the organism back to a state of hedonic neutrality.

Opponent-Process Theory and Separation. Solomon and Corbit's theory of acquired motivation uses the dynamics of drug addiction as a model for the acquisition of many forms of complex motivation, including thrill-seeking sports, imprinting in precocial birds, and love–attachment relationships in humans and monkeys. Rather than seeing drug addiction as an anomalous form of acquired motivation, as have most psychologists, Solomon and Corbit propose that, given some initial exposure to an addicting drug, the addictive cycle which a person slips into results from the normal actions of a healthy nervous system. From a functional standpoint, the response of grief, depression, or despair exhibited by a person upon separation from an attachment object is analogous to the withdrawal symptoms of a heroin addict when he/she is "separated" from his drug. This approach, like Bowlby's, emphasizes that there is a universal form of reaction to separation. Unlike Bowlby, however, Solomon and Corbit see this reaction as a typical aspect of many acquired motivational states.

Details of Opponent-Process Theory: Assumptions and Dynamics of the Model. We shall use the example of drug addiction to exemplify Solomon and Corbit's model because it is a particularly clear-cut prototype of an opponent-process system. First, they assume that there are brain systems whose sole function is to bring the individual back to hedonic neutrality when nearly any intense affect-arousing stimulus or event (pleasant or unpleasant) has occurred. In the course of returning the individual to hedonic neutrality, these systems also produce affective–motivational states of their own which are characterized by a hedonic quality somehow opposite to that of the original affect. Thus, for example, when a drug like heroin stimulates an euphoric "high" (*a* process), a second system automatically comes into play, through central feedback mechanisms, whose sole function is to bring the person back to hedonic neutrality (*b* process). This second opponent-process (*b* process), generated by feedback from the onset of the *a* process and hence a slave process, is assumed to be sluggish in latency and slow to die out or decay. Hence, although the *b* process starts to exert its function while the stimulus or event

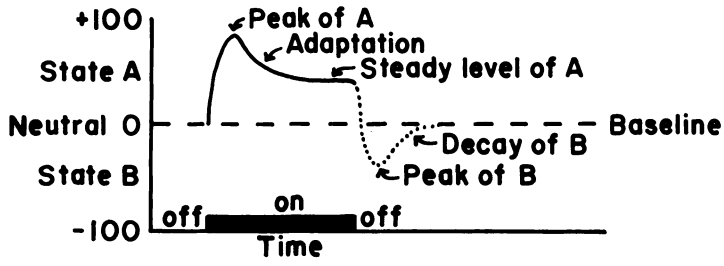


Fig. 4. The manifest temporal dynamics of affect generated by the opponent-process system during the first few stimulations. Features of the affective response state are labeled: the peak of the primary affective reaction, the adaptation phase, the steady level, the peak of the affective afterreaction, and finally the decay of the afterreaction. (The heavy black bar represents the time during which the affect-arousing stimulus is present. The ordinate represents two hedonic scales, each departing from neutrality, one for the primary afeect, the other for the affective after-reaction.) (Adapted from Solomon and Corbit, 1974.)

causing the *a* process is still present, it will not itself be manifest in its full intensity until the stimulus is gone from the body. The onset and offset characteristics of these *a* and *b* processes are depicted in Figure 4.

The *a* and *b* processes are assumed to sum algebraically, the *a* process being positive and the *b* process negative. The result is the observed hedonic behavioral *A* and *B* states seen in Figure 5. In the case of heroin addiction, the initial "rush" or "high" experienced right after taking the drug (peak of the *A* state) is a manifestation of the pure *a* process. However, completely automatically, the *b* process soon begins to subtract from the intensity of the *a* process, resulting in a milder, steady state of pleasure (*A* state: *a*-*b*), which is not nearly as intense as the initial high. When the drug disappears from the body, the *b* process manifests itself in its full intensity for the first time and is generally characterized as a mildly unpleasant craving (*B* state) that gradually dies out.

The next critical assumption of opponent-process theory is that the *b* process gets stronger with repeated use. Its latency to recruit gets faster and it becomes more intense and more long-lasting. In contrast, the *a* process, like any strong unconditioned response, maintains the same strength with repeated use. However, because the *b* process gets strengthened and subtracts in intensity from the *a* process, the resulting *A* state (*a*-*b*) is considerably different from that characterized above. In the case of heroin addiction, after repeated doses of heroin, the initial "high" becomes much less intense and the steady state may not be very different

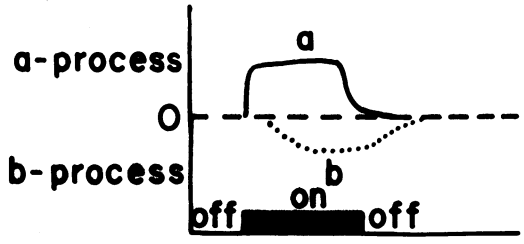


Fig. 5. The underlying opponent-processes which sum to yield the manifest affective response during the first few stimulations. (Depicted in Figure 4.) (Adapted from Solomon and Corbit, 1974.)

from the individual's normal feelings before the drug was ever taken. Furthermore, once the drug has been metabolized the *b* process again manifests itself in its full intensity, and is now characterized by intense craving or abstinence agony which lasts for a long time (*B* state). These changes in the processes and resulting hedonic/behavioral states are illustrated in Figure 6. At this time the person taking heroin has developed a tolerance for the drug (he/she needs more and more of the drug to experience smaller and smaller "highs"). The individual is now in "the addictive cycle," in which he/she constantly anticipates the *B* state and so craves the drug not primarily because of the *A* state it creates (now very weak) but rather because it prevents the *B* state.

It is further assumed that both *A* and *B* states, like any powerful hedonic states, are conditionable to neutral cues. Thus, for example, conditioned stimuli (CSs) such as the "injection ritual" acquire the capacity to elicit the *A* state by virtue of having been paired many times with that state. This is probably the basis for the well-known placebo effect used by many addicts when unable to obtain their usual dose. Other CSs which are paired with the *B* state acquire the capacity to elicit withdrawal symptoms in and of themselves (cf. Wikler, 1965, for a demonstration of this phenomenon in rats).

OTHER EXAMPLES OF OPPONENT-PROCESSES. As a model of drug addiction, opponent process theory provides an interesting framework in which to study the temporal dynamics of the addictive process and the conditioning factors now acknowledged to be so important in relapse following withdrawal (e.g., Teasdale, 1973). Drug addiction, however, is not presumed to be an isolated or unique example of how opponent-process systems are involved in the acquisition of many complex acquired motives. For example, imprinting in ducklings is an acquired motivation that can be profitably studied in an opponent-process framework (see Hoffman and Solomon, 1974; Starr, 1976, 1978, for a more complete discussion of this theory of imprinting). For a young duckling, the presentation of a mother duck elicits a pleasurable state that can be measured by the strength of following behavior and the disappearance of all signs of fear or distress (*A* state: following). The first time the mother duck is removed, the duckling becomes distressed as indexed by a distinctive form of cries (*B* state: distress cries). With repeated exposure to the attachment object, the response to separation becomes more intense and more long-lasting, as indexed by a more prolonged distress crying phase. In opponent

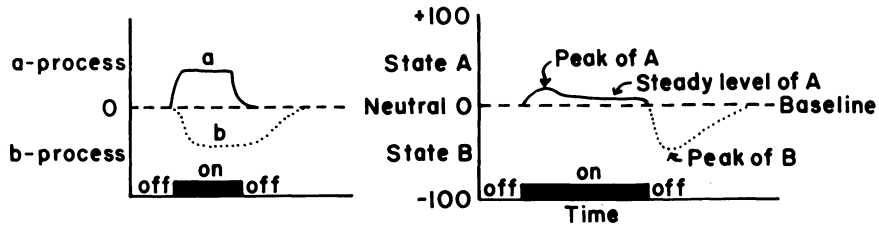


Fig. 6. The left panel depicts the underlying opponent processes after many stimulations: *a* is unchanged while *b* has increased in intensity and duration, and decreased in latency of onset. The right panel depicts the manifest temporal dynamics of affect generated by these opponent processes after many repeated stimulations. (Adapted from Solomon and Corbit, 1974.)

process terms, this is the manifestation of an intensification of the *b* process that comes with repeated exposure. Unfortunately, there is as yet no conclusive evidence that the *A* state is being dampened concurrently with the growth of the *B* state as would be expected if the *b* process were growing. In this view, one measures the strength of attachment of a baby duckling for its mother by how it responds upon separation, just as one measures the strength of an addict's dependence on a drug by how intense the withdrawal symptoms are when the drug is no longer present in the body. Such a characterization of imprinting differs in emphasis from other current theories, in that most attention is directed toward what happens to the duckling when the mother is gone rather than when she is present, and how this changes over time. (See Solomon, 1977; Starr, 1976, 1978, for a discussion of some of the unique predictions of the opponent-process approach to imprinting.)

Although imprinting was once thought to be a unique form of learning that occurs only in precocial birds, it is now more commonly thought to be closely related to the development of attachment bonds in other species (Immelmann and Suomi, 1981). Thus, for example, both Bowlby (1969) and Hoffman and Ratner (1973) have argued that imprinting attachments should be looked at as being on a continuum with attachments in primates and man with respect to such characteristics as speed of formation of the attachment, the length of the sensitive period, and irreversibility of the attachment. In all cases, prolonged and repeated exposure to an attachment object results in the strengthening of a *b* process whose function is to bring the organism back to hedonic neutrality. After prolonged exposure to the attachment object, the attachment should appear less intense (dampening of the *A* state by the growing *b* process), but the potential for distress upon separation from the attachment object greatly increases (*B* state very intense and prolonged).

SEPARATION PHENOMENA IN MONKEYS AS VIEWED FROM AN OPPONENT-PROCESS PERSPECTIVE. Opponent-process theory can be useful in understanding certain changes in patterns of responding that occur in rhesus monkeys during repeated separations and reunions. As described earlier (pp. 555–556) we have discovered a striking pattern of changes in response to separation and reunion in monkeys who have experienced 16–24 4-day separations in their second or fourth year of life. In the first of the two studies, monkeys living in three groups of four each received three series of eight short separations each (4 days apart, 3 days together), beginning when subjects were 16 months of age. After each series of eight separations all three groups had an 8–12-week period of reunion not interrupted by any separation. Four major findings which fit nicely into an opponent-process framework emerged from this study. (See Mineka *et al.*, 1981, for a more complete discussion of some of these results.)

1. The protest phase that initially occurred upon separation gradually disappeared across each series of eight separations. This is illustrated in Figure 7. There are two possible explanations for this finding. First, if protest is conceived of as a frustration reaction that occurs when there is any major disruption in the environment or when there is a violation of expectations (Amsel, 1967), then one would expect such a reaction to habituate as do other frustration reactions when the expectations change through repeated elicitation. Although this first explanation is outside the scope of opponent-process theory, it is not inconsistent with the

theory. Alternatively, the protest reaction may be the first signs of the *B* state (despair). With the repeated exercise of the *b* process that occurs over repeated separations, there may well be qualitative changes in the *B* state that occur upon separation, including a bypassing of the initial state of protest and a faster drop into the despair state. (See Solomon and Corbit, 1974, for a description of other *B* states that change qualitatively with repeated exposure.)

2. In nearly all subjects the intensity of the despair reaction that occurred upon separation became somewhat more severe with repeated separations (Figures 7 and 8). This follows the prediction of the opponent-process model that there should be an intensification of the *B* state due to a growing *b* process with repeated separations. Actually these changes in the despair response are particularly striking in the light of some developmental changes occurring during the second year that mediate against seeing such an intensification. Specifically, the behaviors that best characterize despair (self-clasping and passivity) become less and less common in

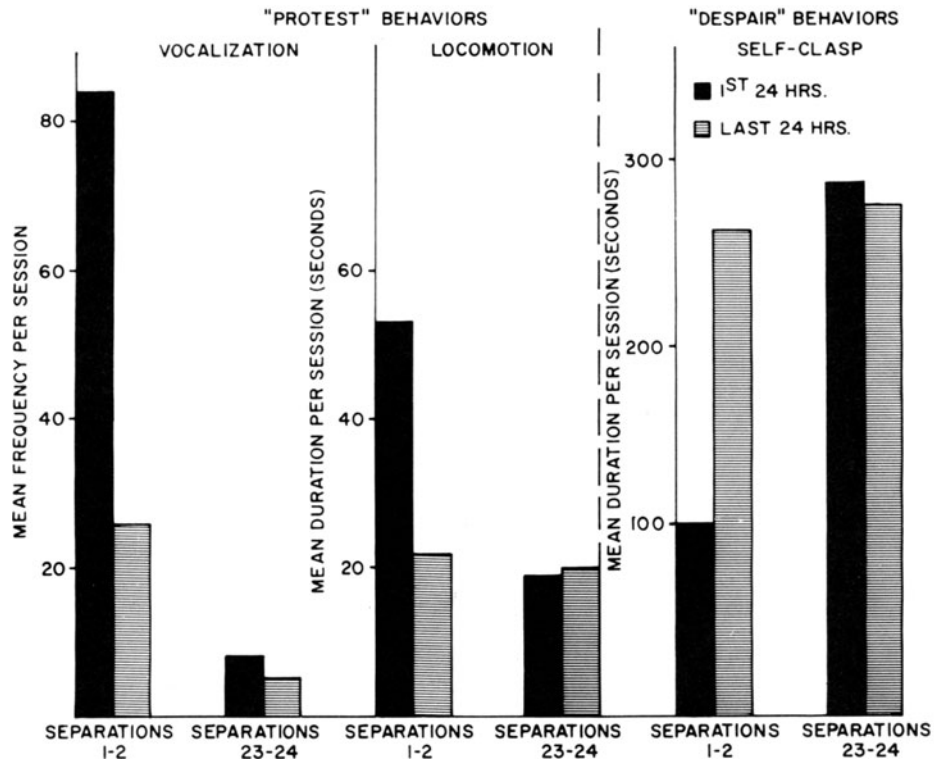


Fig. 7. Changes in levels of vocalization, locomotion, and self-clasp behaviors from the first 24 hr to the last 24 hr of a 4-day period of separation, averaged over the first two and the last two separations, respectively, of a sequence of 24 4-day separations. Vocalization and locomotion, characteristic of "protest," dropped considerably from the first to the last 24 hr within the first two separations, but remained at low levels throughout the final two separations. Conversely, self-clasp, characteristic of "despair," increased considerably within the first two separations from the first to the last 24 hr, but remained at high levels within the final two separations. These data thus demonstrate that with repeated separations, protest behaviors diminished significantly, and despair behaviors began to occur at full intensity immediately upon separation.

nonseparated monkeys over the course of the second year, presumably due to maturational factors (Suomi and Harlow, 1975). Such developmental changes obviously mediate against the possibility of observing increases in such behaviors with repeated separations.

3. The response to reunion (*A* state) diminished as the study progressed in almost all of the subjects. In fact, after many separations, these monkeys appeared to be in a state of chronic despair. Unlike normal monkeys of this age, they tended to sit passively next to one another, self-clasping at high levels, and only rarely engaging in any play or sexual behavior. In opponent-process terms, the *b* process has become so intense that, when subtracted from the *a* process, there is little sign of an *A* state (i.e., normal attachment behaviors are gone). Even during the 8–12-week recovery periods between each series of separations, there was very little sign of recovery.

4. Because of the sluggish onset of the *b* process when *A* is stimulated, opponent-process theory predicts dynamic changes in the intensity of the *A* state within any single reunion period (see Figure 5). In particular, the *A* state should be at its peak intensity immediately following reunion, and diminish gradually as the *b* process recruits and begins to subtract from it in intensity. In line with this prediction there was a definite change in behavior from the first to the last day of reunion;

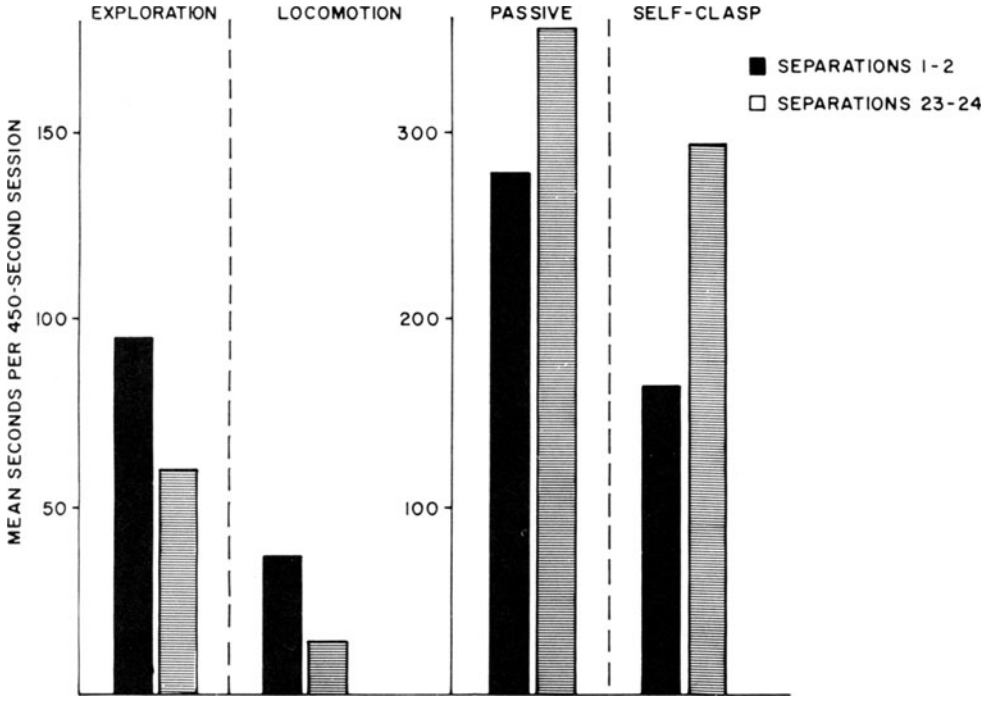


Fig. 8. Levels of exploration, locomotion, passive, and self-clasp behaviors recorded during the first two and the final two separations of a sequence of 24 4-day peer separations. Inasmuch as high levels of exploration and locomotion tend to be indicative of normal activity, while high levels of passive and self-clasp behavior tend to characterize depressivelike behavior, these data indicate that reactions to separation became considerably more severe across the present series of 24 separations.

the monkeys showed less attachment to each other across the 3-day reunion period. In particular, most monkeys exhibited (a) more social behavior at the beginning of the reunion than at the end and (b) more exploratory behavior and locomotion (nonsocial behaviors) at the end than at the beginning of reunion. Overall, these results indicate that the signs of social attachment were strongest at the start of reunion and diminished across the 3-day reunion period. At the end of reunion most of the monkeys' activity was nonsocial in nature, indicating little sign of social attachment behaviors that characterize normal monkeys of this age. Yet, consistent with opponent-process predictions, this was the very time at which separation reactions were very severe.

Recently, we have seen a similar pattern of changes in response to separation and reunion in a study involving older monkeys undergoing repeated separations during their fourth year of life (age 3–4). Some of these monkeys (*S* monkeys) had an extensive previous history of separations during the first months of life, whereas the others (*T* monkeys) had lived together since infancy. Prior to the current series of separations, all *S* and *T* monkeys were housed together during their third year. As in the previous study, four main findings emerged in this study that are consistent with predictions from opponent-process theory. (1) Although in these older monkeys there was very little sign of protest upon separation, what protest that did occur habituated very rapidly after several separations. (2) These monkeys (especially *T* monkeys) showed an even more marked tendency for the despair phase to intensify across repeated separations than did the previous groups described above. Both *S* and *T* monkeys exhibited dramatic increases in self-clasping behavior across the two series of separations. *T* monkeys, in addition, displayed an increase in huddling behavior and a decrease in explorations and locomotion across time. These latter trends were somewhat less marked for *S* monkeys, but this may be partially due to a ceiling effect. Overall, one is left with the impression that these 3–4-yr-old monkeys were experiencing very severe despair upon separation. That the *T* monkeys who had no previous history of separation showed such an intense response may seem surprising at first glance, given the previous reports of little evidence of adverse reactions to separation in monkeys this age (e.g., Bowden and McKinney, 1972). In all previous studies, however, the older monkeys had been separated from groups formed after the first year or two of life. Therefore, their attachments before separation may well have been much less intense than those of the *T* monkeys in this current study, who have been raised together since infancy (see also Kraemer and McKinney, 1979). (3) In addition to the finding that despair became more intense with repeated separations, there was a tendency for the response to reunion to become attentuated across repeated reunions. In particular, there was an overall decline in socially directed behaviors, especially on the third day of reunion. (4) There was also a marked tendency for the peak of socially directed behaviors to occur on the first day of reunion. In both groups the last reunion day increasingly was occupied by solitary environmental exploration and passivity as the study progressed.

The results of these two studies fit nicely into an opponent-process framework. Unfortunately, many of the predictions of the theory cannot be directly tested until further studies are done with the opponent-process framework in mind. For

example, the intensity of the pure *A* state was probably not accurately reflected in these results because the first reunion observations were not made until 15–30 min after reunion occurred, at which point the real peak of the *A* state may have already diminished. To get an accurate picture of the dynamic changes in *A* over time reunion observations should be made immediately following reunion.

In addition, our studies to date do not allow us to tease apart those findings that stem from the operation of purely automatic opponent processes and those that stem from the operation of associative processes in interaction with the opponent processes. For example, in these studies, the same caretakers and experimenters separated the monkeys, reunited the monkeys, and made behavioral observations during both separation and reunion. Thus, potentially powerful conditioned stimuli for the *A* and *B* states—CS_As and CS_Bs—were confounded. In addition, if an observer had become a powerful CS for the *B* state, then the fact that CS_B was present during reunion observations (*A* state) prevents assessing the extent to which the dampened *A* state was the result of associative (CS_B) or nonassociative (strengthened *b* process) factors. Future studies will be designed to differentiate the relative roles of associative and purely automatic opponent processes.

The opponet-process approach to attachment and separation is consistent with most of the data discussed previously regarding variables influencing attachment and separation behaviors. For example, consider the secure base effects discussed earlier (p. 567) (Bowlby, 1969, 1973; Harlow and Zimmermann, 1959). If one assumes that pleasurable and aversive *A* states coexisting in time sum, then one would predict that when an attachment object induces a pleasurable or positive *A* state, then the aversive or negative *A* state induced by any frightening stimulus presented in the presence of that attachment object should be substantially reduced in intensity. Conversely, a frightening stimulus presented just after removal of the attachment object should produce greater than normal distress because of the summation of an aversive *A* state with an aversive *B* state (induced by separation). The theory also predicts that the degree of modulation of the distress reaction should be a function of what stage of the *A* or *B* state the organism is in. So for example, if an organism has just been reunited with its attachment object (peak of *A*), distress reduction should be greater than if the attachment object has been present for quite some time (steady state of *A*). Conversely, a fearful stimulus presented right after separation (peak of *B*) should be considerably more frightening than one presented a long time after separation has taken place. Solomon (1977) has recently presented data from chicks consistent with this prediction. Results from his laboratory indicate that the duration of tonic immobility in chicks is greatly influenced by the presence or absence (decreased and increased, respectively) of an imprinting object (secure base). Furthermore, the extent of the modulation of the immobolity reaction is a perfect function of what stage of the *A* or *B* state the chick is in, that is, maximal reduction of the immobility reaction occurs right after the imprinting object is presented and maximal potentiation of the immobility reaction occurs right after the imprinting object is withdrawn. Unfortunately, we do not as yet know whether analogous temporal dynamics of secure base effects in monkeys exist, but opponent-process theory clearly provides us with a rationale for looking for such effects.

Overall, we feel that the opponent-process approach to attachment and sep-

aration provides one predictive framework in which to pursue the study of the temporal dynamics of attachment and separation. As we have seen, there are at least two special advantages that the opponent-process approach brings to this area not afforded by Bowlby's, Seligman's, or Kaufman's approaches. First, the model makes the most specific predictions about the changes in response to separation and reunion that will occur over time as an attachment develops. This stems from the fact that it is specifically designed to be a model of the temporal dynamics of affect. Second, the model provides a useful framework within which to evaluate associative and nonassociative factors that contribute to the dynamics of attachment and separation responses. Although Bowlby recognizes the importance of both factors, his approach provides no real framework in which to begin to tease apart the relative contributions of each. Seligman's learned helplessness approach does not address such issues since it is basically concerned with determining the essential aspect of the separation state that produces such a profound despair–depression response, that is, loss of control over important life events. Similarly, Kaufman's approach does not address these issues.

Conclusions

We have reviewed the data regarding the reactions of monkeys to separation from objects of social attachment. To date the majority of studies in this area have analyzed the phenomena behaviorally, but recently a number of investigators have begun to study separation-related phenomena from other motivational perspectives and levels of analysis. For example, we are beginning to get a picture of the physiological underpinnings of the affective-motivational states created by separation and hopefully will soon begin to understand how similar they are to other stress-induced states. In addition, a recent cognitive theory—the learned helplessness theory of depression—provides a framework in which to study the cognitive changes that may well precede and determine, or at least accompany, many of the affective–motivational changes that follow separation from an attachment object.

We have also discussed several evolutionary-functional analyses of the separation phenomenon. Those of Bowlby and of Kaufman provide interesting hypotheses as to the origins and functions that the protest–despair response serves, as well as suggesting various factors that may either attenuate or exaggerate this response. Solomon and Corbit's opponent-process approach also provides an elegant predictive framework in which to study both the temporal changes in separation-related affect and motivation that occur across time, as well as the relative contribution to these changes of associative as opposed to nonassociative, purely automatic opponent-process variables.

Thus, the reactions of monkeys to social separation can be profitably viewed from several different motivational approaches. In general, these approaches are not mutually exclusive but rather complement one another. We are confident that all these different perspectives will contribute substantially to furthering an understanding of this complex affective-motivational state in the years to come.

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